

SYSTEM CONFIGURATION REPORT -- USER VS. WORKSTATION IDENTIFIER LOGGING

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Distribution	Harbormaster; Audit Liaison; Finance Manager
Purpose	Inventory of harbor system screens, distinguishing screens that write a user identifier from those that write only a workstation identifier, in support of audit engagement PE-2025-014

1. SYSTEM OVERVIEW

1.1 The District operates the HarborMaster Billing & Registry System (HBRS), installed in 1998 and successively extended. HBRS comprises the billing engine, the waiting-list and offer module, the skip event register, the vessel particulars register, the moorage agreement file, and the accounts receivable ledger. The District's office network has eleven workstations, of which two are counter terminals (HB-COUNTER-1 and HB-COUNTER-2) and nine are office workstations.

1.2 HBRS was designed for a single-operator office. Its audit trail convention is therefore inconsistent: screens written or modified after 2004 write the logged-in user identifier to each transaction row; the older screens -- including the vessel particulars amendment screen, delivered with the original 1998 build -- write only the workstation identifier and a timestamp.

2. SCREEN INVENTORY

2.1 The table below inventories the operational screens, the identifier each writes to its transaction rows, and the module of record.

Module / screen	Writes	Module of record	Counter use
Waiting-list & offers (WL-11)	user identifier	offers register	daily
Skip event register (SK-12)	user identifier	skip events	daily
Billing engine (BL-01)	user identifier	billing ledger	monthly runs
Moorage agreements (MA-07)	user identifier	agreement file	per signing
Receivables ledger (AR-04)	user identifier	AR ledger	daily
Reporting layer (RP-20)	user identifier	read replica	as needed
Vessel particulars amendment (VP-03)	workstation only (HB-COUNTER-2)	particulars register	daily
Retainer list screen (RT-02, decomm. 2019)	workstation only	archived	none

2.2 As the inventory shows, the vessel particulars amendment screen (module VP-03) is the only operational screen in regular counter use that writes a workstation identifier without a user identifier. The screen permits one concurrent session; all nine counter staff share terminal HB-COUNTER-2 for amendment entry, and every amendment row in the register export therefore reads HB-COUNTER-2.

3. THE AMENDMENT SCREEN IN DETAIL

3.1 Screen VP-03 maintains the vessel particulars register: vessel name, hull identification number, official number, flag, length overall, beam, draft, gross and net tonnage, vessel type, hull material, year built, build location, engine make and horsepower, propulsion type, and owner name and address. Each amendment writes one row per field changed, with the old value, the new value, the workstation identifier, and the timestamp.

3.2 In the audit window (January 2022 through August 2026) the register recorded 2244 amendment rows, all logged to HB-COUNTER-2. Attribution of any row to an individual clerk requires the corroborating handwritten counter day logs, as described in the amendment log review summary (CHD-AL-2025-117).

3.3 The screen does not enforce the supporting-document requirement of Resolution 2022-06 section 4.3. Support is a paper-file control: the certificate or haul-out record must be on file, and the register entry should cross-reference it in the source-document field. The screen accepts a blank source-document field on every field, including length overall.

4. CHANGE REQUEST CR-0412

4.1 CR-0412, opened in 2021, requests user-identifier logging for the legacy amendment screen. The change requires modifying the 1998 screen module and its row layout, which the vendor no longer supports; the work must be done by a contractor maintaining the legacy build. Estimated cost at last quotation (2023): \$38,400 plus regression testing of the billing interface.

4.2 CR-0412 has been deferred in the 2022, 2023, and 2024 budgets and remains unfunded through the audit window. The migration deferral record (CHD-IS-2023-009) documents the deferral decision and its conditions, including the compensating control of the Deputy Harbormaster's monthly review of the register against the counter day logs.

4-A. FIELD-LEVEL LOGGING DETAIL

4-A.1 For each register family, the following identifiers are written to the transaction row. This is the field-level evidence for the screen-level inventory in Section 2.

Register / table	Row key	User id field	Workstation field	Timestamp
Offers register	offer_id	entered_by (login)	workstation	entered_ts
Skip event register	skip_id	entered_by (login)	workstation	entered_ts
Billing ledger	txn_id	posted_by (login)	workstation	post_ts
Agreements	agreement_id	created_by (login)	workstation	created_ts
AR ledger	ar_id	posted_by (login)	workstation	post_ts
Particulars register (amendments)	amendment_id	(none)	workstation (HB-COUNTER-2)	entered_ts

4-A.2 The particulars register is the only table whose row has no user-id column at all. The row layout dates from the 1998 schema and cannot accept a new column without a rebuild of the screen module's record definition, which is why CR-0412 is a code change

and not a configuration change.

4-B. SESSION CONCURRENCY ON VP-03

4-B.1 The amendment screen enforces a single concurrent session: a second login at HB-COUNTER-2 while a session is open is refused with the message 'register locked'. Counter practice is therefore to queue amendments within a session and to note the queue in the day log, which is why the day log shows amendment bursts rather than an even spread through the day.

4-B.2 The lock is session-level, not row-level. There is no record of which fields were touched within a session beyond the register rows themselves; the day-log note is the only contemporaneous human account.

4-C. QUARTERLY ARCHIVE

4-C.1 Under the migration deferral's conditions, the register is exported quarterly to a read-only archive with a control total (row count and hash) recorded on the archive label. The 2022-Q3 archive (first after Resolution 2022-06 took effect) and each quarterly archive since are held by the clerk's office. The audit export of 2244 rows ties to the 2026-Q3 archive control total.

5. ACCESS CONTROL

5.1 Access to HBRS is role-based; the user-workstation access register (user_workstation_access_register_2026-06.xlsx) lists each staff member, role, module access, and workstation. Nine staff hold counter roles and are authorized at HB-COUNTER-2. Access reviews are performed semiannually; the last review was June 2026 with no exceptions.

5.2 The auditor read-only account (role READ_ONLY) is logged with the user identifier on every screen, including the legacy screens, because read access passes through the reporting layer added in 2009.

6. CONCLUSIONS

6.1 Screens that write a user identifier: waiting-list and offers, skip event register, billing engine, moorage agreements, receivables ledger, reporting layer (all post-2004 builds).

6.2 Screens that write only a workstation identifier: the vessel particulars amendment screen (VP-03, 1998 build), and the archived retainer-list screen (decommissioned 2019).

6.3 The District's exposure is confined to amendment attribution and is mitigated, though not eliminated, by the handwritten counter day logs and the monthly supervisory review. Funding CR-0412 is the definitive remedy.

7. INCIDENT AND CHANGE HISTORY

7.1 The harbor system's change log for the window shows 34 changes: 29 routine (patches, report tweaks, rate-table updates for Resolution 2022-06 and the annual fee letters), 4 emergency (two NAS fan failures, one blocked guest-moorage posting after a power sag, one failed restore test traced to a corrupt backup set), and 1 deferred indefinitely (CR-0412). No change touched the amendment module; the deferral conditions forbid it.

7.2 Two incidents bear on the audit's data integrity questions. The May 2024 restore-test failure led to the weekly-set verification now in the backup memo (CHD-IS-2024-017); the register archive was unaffected. The February 2023 power sag corrupted one billing batch's first posting; the batch was re-posted from the engine's journal the same night and the reconciliation shows the correction as a reversal and re-entry pair, both rows present in the export.

7.3 No security incident affecting the register is recorded in the window. The counter VLAN isolation of the legacy module was tested by the 2024 network scan; the amendment screen has no network path beyond the counter segment.